Hangman Word Guess Game in Python

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BATCH : 03

1 AIM

To develop a Hangman word guessing game using Python that demonstrates the use of strings, lists, loops, conditional statements, functions, and random module.

2 PROCEDURE

1. Start the program.

2. Import the random module.

3. Create a function to choose a random word from a predefined list.

4. Create a function to display the word with guessed letters and blanks.

5. Initialize an empty list to store guessed letters.

6. Set the number of attempts to 6.

7. Prompt the user to enter a letter.

8. Check if the letter is already guessed; if yes, notify the user.

9. Check if the guessed letter is in the word and update attempts accordingly.

10. Repeat the process until the user guesses the word or runs out of attempts, then display the result.

3 CODE

```
import random

def choose_word():
    words = ["guru", "tarun", "siva", "sridhar"]
    return random.choice(words)

def display_word(word, guessed_letters):
    result = ""
    for letter in word:
        if letter in guessed_letters:
            result += letter + " "
        else:
            result += "_ "
    return result

def hangman():
    word = choose_word()
    guessed_letters = []
    attempts = 6

    print("Welcome to Hangman Game!")

    while attempts > 0:
        print("\nWord:", display_word(word, guessed_letters))
        guess = input("Enter a letter: ").lower()

        if guess in guessed_letters:
            print("Already guessed!")
            continue

        guessed_letters.append(guess)

        if guess in word:
            print("Correct!")
        else:
            attempts -= 1
            print("Wrong! Attempts left:", attempts)

        if all(letter in guessed_letters for letter in word):
            print("\nYou won! The word is:", word)
            return

    print("\nGame Over! The word was:", word)
```

hangman()

4 OUTPUT

Welcome to Hangman Game!

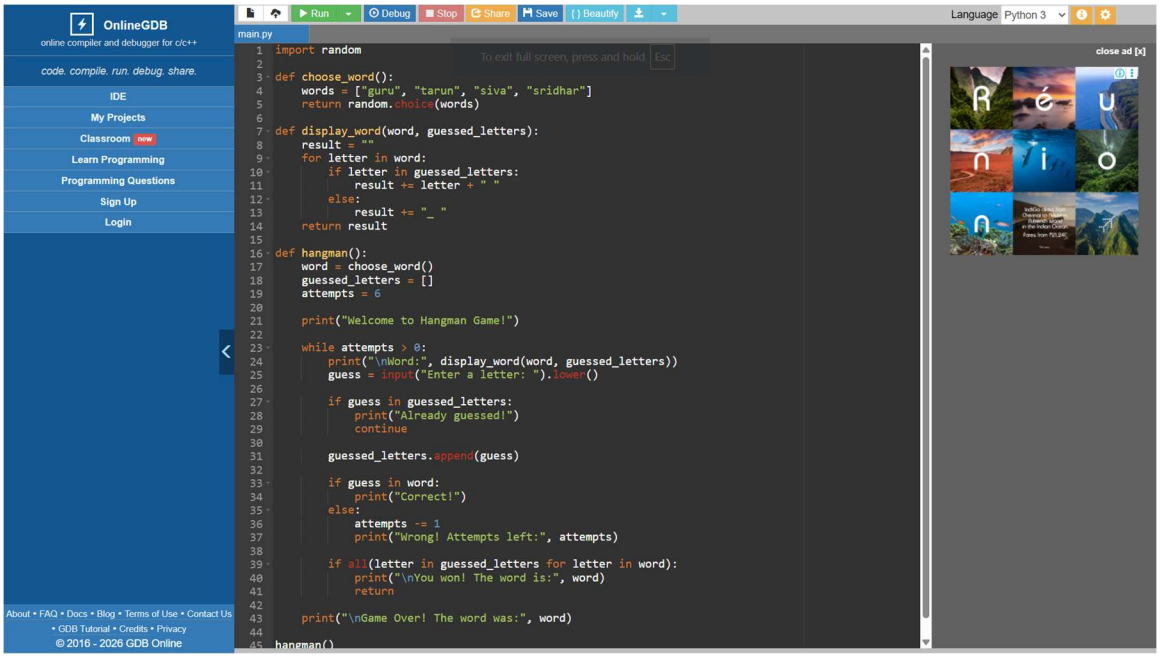
Word: _ _ _ _
Enter a letter: g
Correct!

Word: g _ _ _
Enter a letter: u
Correct!

Word: g u _ u
Enter a letter: r
Correct!

Word: g u r u

You won! The word is: guru



5 RESULT

Thus, the Hangman word guessing game was successfully developed and executed using Python.